

Optimizing Hindi-to-English Translation: Adapting mBART for Resource Efficiency

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Abstract— With the increasing demand for efficient task-specific adaptation of large models in computationally constrained environments, the research explores methods to optimize translation quality while minimizing computational overhead. This work focuses on addressing the challenges of Hindi-to-English machine translation (MT) with a focus on parameter-efficient fine-tuning (PEFT) techniques, namely Low-Rank Adaptation (LoRA) and Quantized LoRA (QLoRA), applied to the pre-trained mBART model. Low-Rank Adaptation trains the layer's parameters which are stores as low-rank decomposition. QLoRA performs quantization of the paramters first, then train the model's layers using LoRA. Utilizing the pre-trained mBART model as a foundation, fine-tuned it with a curated dataset comprising 9,900 diverse Hindi-English sentence pairs, categorized across linguistic dimensions such as idiomatic expressions, tenses, and grammatical structures. The experiments reveal that LoRA achieves a 66.12% reduction in GPU memory usage, a 70.01% reduction in training time compared to full fine-tuning, and 8.16% reduction in inference time, while maintaining comparable METEOR and ChrF scores except for Idiomatic sentences where performance declined. QLoRA, leveraging 4-bit quantization and innovative memory optimizations, demonstrated a 90.06% reduction in GPU memory usage. These results underline the potential of PEFT techniques, applied to pre-trained multilingual models like mBART, in creating scalable, efficient MT systems capable of running on resource-constrained hardware.

Index Terms—Hindi-English Translation, Language Adaptation, Machine Translation (MT), Pre-trained mBART Model.

I. INTRODUCTION

MT is where computer-based systems are used to convert text from one language to another. These systems can function entirely independently (fully automated) or in combination with human assistance (such as post-editing or pre-editing by human translators). Increasingly, there is more and more information being produced that makes it impossible to meet the demand through having subject matter experts. Which means there is an increasing demand for information access tools for various audiences. Its emergence as a potential solution to this

problem has made it a hot topic in both research and professional settings [1]. Real-time translation in online communication systems, cross-language business communication [2], cross-lingual retrieval [3], and supporting clinical communication [4] by eliminating the language gap between healthcare providers and patients are just a few examples of what it could do.

With developments in Artificial Intelligence (AI), MT has become better than in the past decades. Improvements in translation accuracy, fluency, and contextual understanding provide an ever-more reliable and scalable AI-driven MT solution in a world where the demand for effective multilingual communication keeps growing [5]. Pre-training methods, where the layers of a model are new i.e. untrained, needs a huge amount of data which can be a slight constraint for adapting a machine translation system. As a means to counteract this issue, research has resorted to employing transfer learning techniques. Enhancing models with existing knowledge through transfer learning helps to work with only small amounts of data by supporting that knowledge learned by a model trained on a more general task applies to a specific one. As an example, authors successfully classified mammographic tumors with transfer learning on only 607 full-field digital mammographic images, showcasing the capability of this approach to reach a high-performance when only little data is available [6]. Transfer learning has been successfully used for classifying images of macular degeneration and diabetic retinopathy in other studies [7]. But it is challenging to decide which layers of the model will be re-trained and which will be frozen which is crucial to ensure that the performance of the model is optimized for the desired task without obliterating information learned by the model during initial model training [8]. Deep learning models are considered the backbone of MT systems. However, Deep learning based MT (usually hosted in cloud) has latency issues and stresses the network. Furthermore, continual adaptation and the need for privacy in sensitive applications like healthcare make cloud-based MT solutions even more challenging. Similar to the challenges faced by MT systems, large language models (LLMs) also have deployment problems. This relates to both their energy demands which have environmental consequences and also the expanse they are vulnerable to adversarial attacks and then, their slow real-time responsiveness making them less useful for chatbots and translation services [9]. Three key phases of ML/DL model deployment, device cost, model performance, and data privacy, affect the deployment in companies developing software-intensive embedded systems according to a multi-case study [10] of seven companies. These problems can be solved with Edge AI method, where data would be processed

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in proximity to its source, thus, reducing the time taken to communicate and hence improving responsiveness in translating [11]. For edge servers, the predominant literature around pioneering research is the physical resource obstacles that a huge and complex model like mBART (and all the other most fashionable NMT frameworks) imposes. Another challenge is different computing environments of mobile and embedded devices which requires to design specific models and optimization methods to be able to exploit mobile and embedded hardware capabilities [12]. In cases where processing is to be done at the same time, models need to be optimized to run without getting constrained by available hardware [13]. When fine-tuning these large models to improve personalization performance, resource demand is only further compounded, necessitating efficient strategies like LoRA [14] and QLoRA [15]. In addition, domain-specific data is also cumbersome to acquire: it is heterogeneous, messy, and noisy, rendering collection and filtering a laborious process in contrast to the acquisition of common-sense knowledge. As a result, these problems not only bring unnecessary training redundancies but raise cost of labor, and therefore worsen the deployment of functional personalization systems [16]. With Hindi spoken by 43.6% of the population of India alone, in addition to its prevalence in the rest of the world [17], advancing machine translation for this language pair is of utmost necessity for better communication globally and overcoming linguistic challenges. Hybrid Machine Translation (HBMt) is also an approach which combines rule-based MT (RBMT) methods with statistical (SMT) or neural (NMT) models, leveraging both linguistic rules and data-driven approaches to improve translation accuracy [18]. But, recent advancements highlight that Transformers outperform NMT models and avoid issues like vanishing gradients, making them more effective for translation tasks, achieving better performance [19]. This is the reason a transformer based model, mBART, was used for this research.

However, there are significant gaps in research in the application of PEFT in Hindi-to-English machine translation (MT) which will be addressed. A significant problem is that low resource languages such as Hindi are not as well represented in the literature as common language pairs such as English-French and English-German. While Hindi is among the most common languages in the world, it is low-resourced which means a dedicated datasets for idioms is not available. Due to this limitation, PEFT techniques are not utilized to their full potential while translating Hindi. LoRA and QLoRA have enabled a significant reduction in fine-tuning overhead, however little attention for Hindi-English translation has been received, with most studies so far focusing on NLP tasks like text classification [20] and summarization [21]. The syntactic disparity between its SOV (Subject-Object-Verb) structure and that of English, which is SVO (Subject-Verb-Object) also make it difficult to obtain high quality translations. While a number of multilingual pre-trained models, e.g., mBART [22] and mT5 [23] have been used for the Hindi-English MT task, the exploration of applying PEFT techniques like LoRA, and QLoRA in particular for these models is limited. This paper aims to narrow gaps by optimizing the balance between translation, memory allocation, time utilization during training

and the inference time with the help of PEFT methods for Hindi-to-English translation. This work could significantly contribute to the machine translation of various other language pairs. Additionally, it may provide valuable insights to researchers on the benefits of these recently developed PEFT techniques in the context of machine translation. The transformer-based mBART model was fine-tuned using these techniques for Hindi-English translation. A varied Hindi-English parallel corpus was constructed, consisting of sentence pairs spanning multiple linguistic dimensions, varying degrees of grammatical complexity, like simple vs. compound, or active vs. passive voice, and direct vs. indirect speech, as well as phrases with idiomatic use were included. Further, the dataset covered diverse contextual domains such as transliteration, news articles, technical content, and sentences across different tenses.

The rest of the work is structured as follows. Section 2 provides a literature overview which outlines key studies. Section 3 describes the research methodology. Results are discussed in Section 4, which presents the quantitative insights and the result analysis with the details of experiments conducted towards final result is discussed in Section 5. Section 6 concludes by discussing the implications of the findings, research limitations and future study recommendations.

II. LITERATURE REVIEW

This section is about literature that connects Parameter-Efficient Fine-Tuning (PEFT) with machine translation (MT). AI-based MT systems are a rapidly evolving field of research. It has the aim to improve efficiency and accuracy, especially under constraints such as memory, latency, and training time. The application of PEFT to MT—particularly for low-resource languages like Hindi—remains underexplored. A few key works have started bridging this gap. Ou et al. [24] fine-tuned the Llama2-13B model using NEFT and QLoRA for financial sentiment analysis, achieving 0.8392 accuracy and outperforming RoBERTa and DeBERTa variants. While not MT-specific, this demonstrated QLoRA's domain-specific adaptation capabilities. Zhuohan Li et al. [25] proposed a "Train Large, Then Compress" strategy using large Transformer models on WMT14 English-French translation. They showed that models trained briefly and then heavily compressed not only converged faster but also achieved higher BLEU scores compared to smaller baseline models while offering 60–70% memory reduction and 2–3x faster inference through 6-bit quantization. He et al. [26] presented a fast and accurate NMT model leveraging translation memory (TM) within a Transformer framework. Their model used a single bilingual sentence as TM, optimized with a novel training criterion. On translation tasks such as WMT17 Zh→En, they achieved a BLEU score of 28.3 and reduced training time per epoch by 75% (from 17,841s to 4,518s) while maintaining only an 8.96% parameter increase. Inference time improved 2–4x. Liu et al. [27] addressed MT training inefficiencies using a bi-directional RNN enhanced with a tanh activation function to better integrate historical and future context. For

German→English translation, they attained a BLEU score of 22.77 with a 66% reduction in training time (8.5h vs. 23h) and for English→French, they achieved 18.52 BLEU, surpassing OpenNMT by 3.13 BLEU points. Several works have focused on enhancing model performance under computational constraints, which aligns with PEFT's goal of reduced resource dependency. EdgeTran [28] combined device-aware co-search with surrogate models and BOSHCODE to reduce model size by 2.8x, latency by 15%, and energy usage by 10x compared to BERT-Base. Park et al. [29] introduced techniques like FlashAttention, speculative decoding, and continuous batching to optimize inference on AI accelerators. MeLo [30] demonstrated the effectiveness of LoRA for medical image diagnosis, achieving diagnostic performance on par with full models while using only 0.17% trainable parameters and 0.5MB additional storage per task. Similarly, Zhang et al. [31] proposed an end-edge-cloud framework for AI-generated content (AIGC) services that utilizes federated learning and deep mutual learning to fine-tune large models on distributed edge devices. While initially aimed at medical applications like X-ray diagnosis, the approach improved diagnostic accuracy but suffered performance degradation as more federation nodes were added, highlighting the challenge of scaling PEFT in distributed environments. LoRA for medical image diagnosis, achieving diagnostic performance on par with full models while using only 0.17% trainable parameters and 0.5MB additional storage per task. Across domains, PEFT techniques have shown promise in generalization, latency reduction, and parameter savings. MTLORA [32] applied LoRA in a multi-task learning setting and achieved accuracy improvements over full fine-tuning while reducing trainable parameters by 3.6x. IISAN [33] introduced a decoupled PEFT architecture for multimodal recommendation tasks, achieving 46.74% accuracy with a 97.89% reduction in trainable parameters and 59.59% lower training time. Chakraborty et al. [34] used PEFT for code vulnerability detection and noted significant performance drops (up to 95 percentage points in precision) on real-world datasets, which were partially mitigated with data augmentation (30% improvement). Other relevant works include Maatouk et al. [35], who explored LLM applications in the telecom sector, emphasizing challenges like explainability and instability—key considerations in deploying LLMs for real-time MT. A skip-gram and edge integration approach for English translation was proposed [36] that yielded improvements of 10.79% in accuracy and over 10% in other metrics like matching time and complexity. Bhardwaj et al. [37] provided a comprehensive survey on LLM deployment at the edge, highlighting energy and resource constraints, which align with the goals of memory-efficient MT. Yu et al. [38] reviewed visual tuning techniques for adapting foundation models, discussing generalization trade-offs relevant to cross-domain PEFT tuning.

Recent scholarly works have made noteworthy advances in multilingual and low-resource machine translation. However, there is still progress yet to be made. The EMS model

proposed an effective dual-encoder structure for generating multilingual sentence embeddings, but neglected to address generative translation tasks or parameter-efficient techniques [39]. Alternatively, WDSRL introduced word-level domain-aware representations to boost translation quality, but relied on manually annotated domain labels and complete parameter tuning which limits scalability [40]. Additionally, Li et al. presented the P-Transformer for document-level translation with enhanced position encoding, although its increased memory usage constrains scalability on lengthy documents [41].

While notable advances have been achieved, further work is still needed to develop more generalized, data-efficient, and scalable approaches for machine translation, especially when resources are limited. Several gaps persist, particularly in leveraging advanced fine-tuning and optimization techniques. For example, although parameter-efficient methods like LoRA and QLoRA have demonstrated effectiveness in reducing training overhead and improving model deployment in domains such as finance, medical imaging, and recommendation systems, their targeted application to neural MT tasks—especially involving multilingual models like mBART for Hindi-English translation—has not been sufficiently explored. Further, models like MTLORA and IISAN, while showcasing success in multi-task and multimodal contexts respectively, have not been benchmarked against each other or explored in MT scenarios for tasks requiring idiomatic, domain-specific, or tense-diverse translations. There is also little work evaluating how well LoRA and QLoRA can be interchanged or tuned in low-resource MT settings. Moreover, although some optimization strategies like quantization and edge-aware pruning have been proposed, there is still a lack of comprehensive study on paged optimization or fine-grained memory compression during PEFT processes for MT. Building on these gaps, this research investigates the optimization of pre-trained mBART models using parameter-efficient fine-tuning methods, specifically LoRA and QLoRA. While many PEFT techniques have been evaluated in domains like finance and recommendation systems, few studies compare these approaches directly in the context of machine translation. This work addresses that limitation by conducting controlled comparative experiments between LoRA and QLoRA, exploring hyper-parameter optimization and the trade-offs between memory efficiency, training time, inference time, and translation quality. The findings offer valuable empirical insights that contribute to the development of scalable and efficient multilingual machine translation systems suitable for both academic and industrial use.

III. RESEARCH METHODOLOGY

This study takes a systematic approach to enhance existing translation frameworks demonstrated in Figure 1. It begins with a diverse dataset collection and preprocessing, such as data splitting and tokenization. The mBART base model is taken as a foundation model and is fine-tuned with computational and storage optimization. The resulting mBART fine-tuned model was evaluated on quantitative

metrics — METEOR and ChrF, and some deployment considerations — time to train, response speed and storage needs.

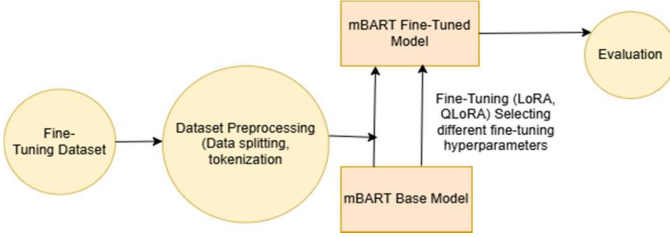


Fig. 1. Research methodology for refining the translation model

A. Data Collection

The dataset is well-structured for Hindi-English translation research and is further subdivided into three major categories, which are: Standard Sentences, Idiomatic Expressions, and Voice and Speech Constructions, with a total of 9,900 sentence pairs. It has multiple sub-categories like sentence type, tense, and so on for each primary category. The Standard Sentences and Idiomatic Expressions are divided into Declarative, Exclamatory, Imperative, Interrogative, Negation, Complex, and Compound sentences, and each sentence type is divided into nine tenses, where there are 50 sentence pairs per tense, so 3,150 pairs for each category. Voice and Speech Constructions include Active Voice, Passive Voice, Direct Speech, and Indirect Speech, each also divided into nine tenses, with 100 sentence pairs per tense (50 technical and 50 news context), totaling 3,600 pairs.

B. Data Preprocessing

The dataset underwent essential preprocessing steps to ensure effective training of the machine learning models and accurate translation results. Initially, the dataset was split into training, validation, and test sets in a 70:15:15 ratio, maintaining a balanced representation of each category and sub-category to ensure the model is trained, validated, and then tested on the data split comprising of each linguistic variation the dataset consisted. Tokenization was performed using the MBart50 Tokenizer.

C. Model Preparation

Following the preprocessing stage, the mbart-large-50-many-to-many-mmt existing model available on Hugging Face has been utilized as the foundational model in this research. This model is a fine-tuned checkpoint of mBART-large-50 [42]. It is based on BART [43], a denoising autoencoder model for language generation tasks. During training, learning rate was set to $1e-4$ and the number of epochs to 5. Full fine-tuning simply means updating all original parameters of a model. Using LoRA, low-rank adapters were trained, allowing to fine-tune the model without modifying the original parameters extensively [14]. LoRA possesses several key advantages. The common part of the model can be frozen, and to change the task the model is required for, matrices A and B in Figure 2, need to be replaced. This reduces the memory needs and task-switching overhead. Training is done faster since computation

of the gradients for most parameters is not needed. It also reduces the memory needs of the model when fine-tuning because the optimizer states and the checkpoints for frozen layers need not be stored [14].

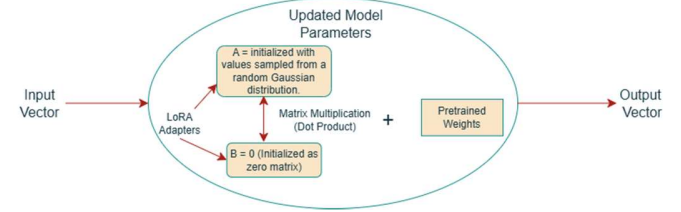


Fig. 2. Reparameterization performed in LoRA technique [14].

QLoRA technique initially quantizes the base model to reduce memory requirements and since the quantized model would lack in accuracy, fine-tuning on task-specific data would be required for which LoRA fine-tuning technique is applied, see Figure 3. QLoRA introduces multiple innovations designed to reduce memory use without sacrificing performance: (1) 4-bit NormalFloat, an optimal quantization data type for normally distributed data that yields better empirical results than 4-bit Integers and 4-bit Floats, (2) Double Quantization, a method that quantizes the quantization constants, (3) Paged Optimizers, to avoid the in-memory surges that occur when processing a large amount of data [15].

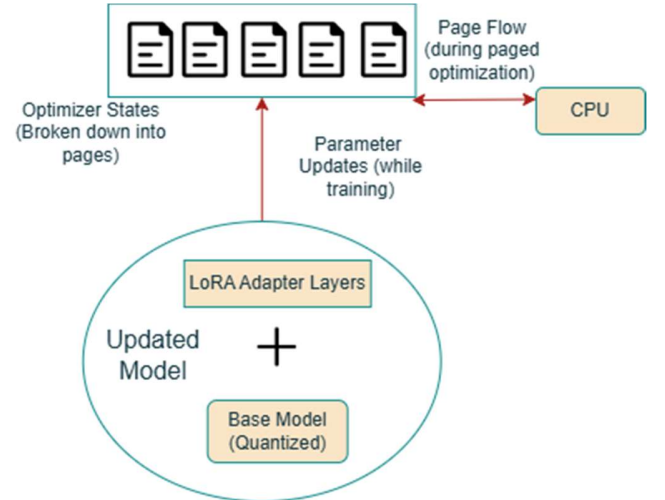


Fig. 3. QLoRA compresses the model, refines the model using LoRA and uses the concept of paging to handle memory spikes, providing better memory efficiency than LoRA [15].

D. Platforms and Frameworks Used

Experiments utilized the Google Colab cloud platform employing NVIDIA T4's with CUDA support, Python code was written in Jupyter Notebook format. Data processing was implemented using pandas library for CSV operations and datasets library was used for seamless handling of data. The torch library was used to run GPU computations in the training pipeline and peft library to implement parameter-efficient fine-tuning techniques. It was required to quantize the model to implement QLoRA, which was done through the BitsAndBytesConfig library. Lastly for the evaluation metrics, we computed METEOR scores using nltk library and chrF

scores using sacrebleu library, measuring the translation quality between Hindi and English.

IV. RESULTS

This section provides an overview of the results. The aim of this work is to show that the optimizations achieve high-quality translations with reduced computational costs. The decreased memory footprint enables the translation model to be run on low-end hardware, which makes deployment possible on edge devices or lower-cost servers. Shorter training times are essential for speeding up experimentation with different model architectures, refreshing models with new patterns of language, and adjusting to changes in language in the real world. Table 1 compares the fine-tuning methods (Full, LoRA, and QLoRA), in terms of GPU memory usage; average training time per epoch; average inference time; and METEOR and ChrF scores on translation. It demonstrates the resource-economical and effective nature of LoRA and QLoRA compared to full fine-tuning. It gives an idea of how each approach trades off performance and computational cost.

TABLE I
COMPARATIVE PERFORMANCE ANALYSIS OF FINE-TUNING APPROACHES: FULL, LoRA, AND QLoRA

Models	Model GPU memory (in MB)	Average training time per epoch (in seconds)	Average METEOR score	Average ChrF score
Full FT	7008.164	887.55	0.895	0.861
LoRA FT	2374.524	266.15	0.859	0.811
QLoRA FT	696.82	343.70	0.845	0.793

V. DISCUSSION

The systematic investigation, conducted analysis of different fine-tuned models with the foundation model and extensive

experiments to optimize the LoRA and QLoRA training configurations. Techniques like LoRA and QLoRA help with resource efficiency significantly. LoRA (Low-Rank Adaptation), with parameters ($r=16$, $\alpha=512$, $\text{dropout}=0.0$, $\text{target_module}=[k_proj, v_proj]$), achieves a 66.12% reduction in GPU memory usage and 70.01% depletion in time to train in contrast with the fully fine-tuned model. QLoRA parameters ($\text{bits} = 4$, $\text{double_quant} = \text{False}$, $\text{compute_dtype} = \text{bfloat16}$, $\text{paged_optimization} = \text{True}$), demonstrates a 90.06% reduction in GPU memory usage and 61.27% reduction in training time over full fine-tuned model.

TABLE II
TRANSLATION PERFORMANCE AND MEMORY REQUIRED BY BASE MODEL

Model GPU memory (in MB)	METEOR score	ChrF score
2331.274	0.646	0.583

The base model was not performing well in Hindi-English translation initially, as show in Table 2, because at the core it lacked the understanding in its pre-trained data. In order to tackle these limitations, the first step was to apply full fine-tuning, which brought a huge gain in translation quality. . However, this improvement came at a substantial computational cost, considering additional GPU memory and training time.

TABLE III
CATEGORY-WISE PERFORMANCE ANALYSIS OF TRAINED AND FOUNDATION MODEL

Category	Base	Full FT	LoRA	QLoRA
Idiomatic	0.466 / 0.387	0.797 / 0.735	0.704 / 0.607	0.669 / 0.567
Standard	0.731 / 0.668	0.932 / 0.910	0.918 / 0.890	0.915 / 0.885
Voice & Speech	0.799 / 0.749	0.956 / 0.938	0.955 / 0.936	0.951 / 0.928

The values in the table are mentioned as - METEOR score / ChrF score.

TABLE IV
TRANSLATION EXAMPLES OF STANDARD CATEGORY SENTENCES

Hindi Source	Base mBART	Full Fine-tuned	LoRA Fine-tuned	QLoRA Fine-tuned
क्या वैज्ञानिकों ने नई खोज की है?	What have scientists discovered?	Have scientists made a new discovery?	Have the scientists made a new discovery.	What have scientists discovered newly?
टीम नई सुविधाओं का परीक्षण करती है।	Tests new facilities.	The team tests new features.	The team tests new features.	The team tests new features.
मुद्दों पर बहस की जा रही होगी।	Issues will be discussed.	Political issues will be being debated.	Issues will be being debated.	The issues will be being discussed.
गौर करें कि आपने सभी समस्याओं का समाधान कर लिया होगा।	Please note that you have solved all the problems.	Ensure you will have solved all problems.	Make sure you have resolved all issues.	Make sure you have resolved all the issues.
तुम्हारी कोई गलती नहीं थी।	There was nothing wrong with you.	It was not your fault.	You had made no mistake.	You had made no mistake.
टॉफी बांटो।	Share the toffee.	Distribute candies.	Distribute the toffee.	Spread the toffee.

वह बाल कटा रही है।	She was cutting her hair.	She is getting a haircut.	She is shaving her hair.	She is shaving her hair.
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TABLE V
TRANSLATION EXAMPLES OF VOICE/SPEECH SENTENCES

Hindi Source	Base mBART	Full Fine-tuned	LoRA Fine-tuned	QLoRA Fine-tuned
वह बता रही है कि वेस्ट बैंक में बस्तियों का विस्तार हो रहा है।	He says that the West Bank is expanding its settlements.	She is stating that the windfall is spreading in the bank.	She is reporting that the population is expanding in the West Bank.	She is reporting that the villages in the West Bank are expanding.
बिजली आपूर्ति सिस्टम चालू करती है।	The power supply system is running.	The electricity supply operates the system.	The power supply turns on the system.	The power supply turns on the system.
चिकित्सक ने परीक्षण किया।	The doctor did the examination.	The physician performed a test.	The doctor conducted the test.	The doctor did the test.
वह बता रही है कि प्लेइडस स्टार क्लस्टर स्पष्ट दिख रहा है।	He says that Places Star Cluster is clearly visible.	She is stating that the Pleiades are clearly visible.	She is reporting that the Pleiades star cluster is clearly visible.	She is reporting that the Places Star cluster is clearly visible.
चुनाव में मतदान किया जा रहा था।	The election was being held.	The election was being voted in.	The polling was being conducted in the election.	The vote was being taken in the election.
वे कह रहे हैं कि फ्रांस साइबर हमलों से चिंतित है।	They say France is concerned about cyber attacks.	They are saying that France is troubled by cyber threats.	They are saying that France is concerned about cyberattacks.	They are saying that France is worried about cyberattacks.
मीडिया जनता को जागरूक करती है।	The media informs the public.	The media educates the public.	The media raises public awareness.	The media raises public awareness.
हम खतरे की जानकारी प्राप्त कर रहे हैं।	We're getting information about the threat.	We are receiving danger alerts.	We are receiving threat intelligence.	We are obtaining the threat information.
महिला सशक्तिकरण के कार्यक्रम चलाए जा रहे हैं।	Women empowerment programmes are being implemented.	Women's empowerment programs are being run.	Programmes for women empowerment are being run.	Women empowerment programmes are being run.

TABLE VI
TRANSLATION EXAMPLES OF IDIOMATIC SENTENCES

Hindi Source	Base mBART	Full Fine-tuned	LoRA Fine-tuned	QLoRA Fine-tuned
उन्होंने अपने व्यापार को बढ़ाने के लिए बहुत पापड़ बेले।	He sold a lot of sausages to increase his business.	They went through a lot of trouble to expand their business.	They went through a lot of trouble to expand their business.	They made a lot of trouble to expand their business.
आप तो ईद का चाँद हो गए!	So you have become the moon of Eden.	You have become a rare sight!	You became a rare sight!	So you became the moon of Eden!
उसने कभी ऊंट के मुँह में जीरा नहीं डाला है।	He has never put a whisper in the mouth of a camel.	He has never offered a drop in the ocean.	He has never given a drop in the ocean.	He has never given a cumin seed to a camel.
उसने ईंट का जवाब पत्थर से देकर दुश्मन को करारा सबक सिखाया।	He answered the bricks with stones and taught the enemy the Karara lesson.	He retaliated fiercely and taught the enemy harsh lesson.	He gave a quarrelsome reply and taught the enemy a harsh lesson.	He gave a stone reply and taught the enemy a harsh lesson.
तुम्हें पत्ता-पत्ता पता था।	You knew the leaf and the twig.	You had known a lot.	You knew everything.	You had known very little.

क्या तुम आसमान सिर पर उठा रहे हो?	Do you raise the heaven above your heads?	Are you raising the sky on your head?	Are you making a big fuss?	Are you raising the sky above your head?
वह तिल का ताड़ नहीं बना रही थी।	She was not making a shell.	She was not exaggerating.	She was not making a mountain out of a molehill.	She was not making a mountain out of a molehill.
वह एड़ी-चोटी का जोर लगा रहा था और सफलता पाने की कोशिश कर रहा था।	He was pushing the wheel and trying to succeed.	He was putting in great effort to achieve success.	He was trying his utmost and striving for success.	He was trying hard to achieve success.
वह उनकी आंखों का तारा बना होगा!	It will be a star in their eyes!	He will have been deceived!	He will have become the apple of their eye!	He will have become the apple of their eye!
उसकी लंबाई आसमान से बातें नहीं कर रही थी।	His length was not talking to the sky.	His height was not sky-high.	He was not very tall.	He was not being very tall.
वह उसकी बात सुनकर आग बबूला हो गया था!	He heard his words and the fire was consumed.	He had been very angry hearing his words!	He had become furious after hearing him!	He had become furious after hearing him!
चार दिन की चाँदनी फिर अंधेरी रात हो गई।	Four days of moonlight turned into a dark night.	The good times were short-lived.	After four days, it became dark again.	Four days turned into a dark night.
अब चिड़ियों के चुगने से खेत उजड़ चुका था।	Now the field had been lit up by the birds.	It was too late to make amends.	Now the field had been lit up by bird calls.	Now the field had been burned by birds' strike.
क्या चोर की दाढ़ी में तिनका दिख रहा है?	Is there a tree in the thief's beard?	Is he showing signs of guilt?	Is there a veil in the thief's beard?	Is there a veil in the thief's beard?
तुमने नाचने के लिए घुंघरू बांध लिए थे।	You wore knees to dance.	You prepared for the task.	You had worked hard to dance.	You had worked hard to dance.
उनके विचार उन्नीस-बीस ही थे।	His thoughts were nineteen and twenty.	Their opinions were slightly different.	His ideas were very narrow.	His thoughts ranged from nine to two.

The translation performance of LoRA and QLoRA are comparable to that of full fine-tuning model, as shown in Table 3, in categories- Standard and Voice & Speech sentences. The LoRA and QLoRA are good alternatives of full fine-tuning if we want to improve the model for improving tense translation, word order (as shown in Table 4), and sentences with new terms related to news and technical context (as shown in Table 5). However, the translation performance of LoRA and QLoRA significantly diminishes as compared with full fine-tuning in the case of Idiomatic sentences (as shown by some examples in Table 6, some examples do show good translation, but most of them are not good, which leads to a decrease in translation performance for idioms). This may be because idioms are rare, non-compositional expressions whose meanings cannot be inferred from individual words, making them difficult to adapt with low-rank updates. These methods only fine-tune a small portion of the model while freezing the rest, limiting their ability to deeply alter semantic representations needed for idioms. Additionally, their low-rank constraint restricts expressive power, making it hard to capture the complex mappings idioms require. Since LoRA/QLoRA assumes the base model's knowledge is mostly adequate, they are less effective when the base model already handles idioms poorly.

This shows that prioritizing resource efficiency using these techniques can't provide good translation performance of figurative sentences. There are places where LoRA performs better than full FT because it adds task-specific layers rather than updating the original parameters. Since the base model was pre-trained on different data, modifying its core weights during full fine-tuning may hinder its ability to adapt effectively to task-specific patterns. There are also examples where QLoRA translated less efficiently, this may be because of lossy reconstruction of quantized parameters but the difference in METEOR and ChrF scores aren't huge.

TABLE VII
IMPACT OF LoRA RANK ON MODEL PERFORMANCE AND
RESOURCE UTILIZATION

LoRA Rank (r)	Additional GPU Memory (in MB)	METEOR Score	ChrF Score
2	5.06	0.798	0.732
12	30.38	0.796	0.733
16	56.75	0.802	0.736
48	137.75	0.799	0.734
64	192.43	0.796	0.731

What is LoRA Rank? To understand this, we would require a

little background. Pre-trained original weight matrix is represented as $W_0 \in R^{d \times k}$, where d and k are dimensions of the matrix R , and ΔW is the weight matrix of LoRA adapter layers [14]. The LoRA adapter layer's weights are stored as low-rank decomposed matrices $A \in R^{d \times r}$ and $B \in R^{r \times k}$ [14]. So, rank r is one of the dimensions of decomposed matrices. So, as rank increases the dimensions of the decomposed matrices increase, the amount of information from parameters which is being stored increases, and thus translation quality increases, as shown by METEOR and ChrF scores in Table 7, but the memory usage also increases. However, during forward pass the LoRA layer's weight matrix ΔW is scaled by the factor of α/r , where α is LoRA Alpha parameter, thus increasing the value of r only helps improve the translation quality upto rank=16, after that the quality degrades because the impact of weight in forward pass inference is inversely proportional to the value of rank. Thus, rank=16 was chosen as the sweet spot. The forward pass can be represented as:

$$W_0 \cdot x + \Delta W \cdot x = W_0 \cdot x + B \cdot A \cdot x \cdot \left(\frac{\alpha}{r}\right) \quad (1)$$

where x is the input matrix [14].

TABLE VIII
IMPACT OF ALPHA VALUES ON MODEL PERFORMANCE

Alpha	METEOR Score	ChrF Score
8	0.786	0.720
32	0.799	0.736
64	0.803	0.740
128	0.810	0.748
512	0.820	0.758

Since the alpha value scales the LoRA learnt parameters, as shown in (1), the translation performance of the model increases as alpha value increases with no change in additional memory required, as shown in Table 8, because the alpha parameter doesn't influence the dimensions of decomposed matrices.

TABLE IX
COMPARATIVE ANALYSIS OF DIFFERENT LoRA ATTENTION MODULE COMBINATIONS

Module	Additional GPU Memory (MB)	Avg. Training Time per Epoch (s)	METEOR Score	ChrF Score
[q_proj, v_proj]	43.25	264.69	0.772	0.706
[q_proj, k_proj, v_proj, o_proj]	56.75	279.67	0.828	0.771
[k_proj, v_proj]	43.25	266.15	0.829	0.771
[q_proj, k_proj]	43.25	278.47	0.797	0.735
[v_proj, o_proj]	29.75	246.52	0.715	0.649
[q_proj, k_proj, v_proj, o_proj, fc1, fc2]	101.75	318.95	0.829	0.771

Moreover, the LoRA target_module parameter is also important. It specifies where to insert the LoRA trained adapter layers in a transformer architecture model. Selecting target_module is quite subjective and requires experimentation. For the task of Hindi-to-English language

translation ["k_proj", "v_proj"] proved to be the most beneficial because it gave the best trade-off for computational cost and translation performance, as shown in Table 9.

TABLE X
COMPARISON OF MEMORY USAGE AND TRAINING PERFORMANCE METRICS BETWEEN PAGED AND NON-PAGED OPTIMIZATION

Metric Category	Metric	Paged	Non-Paged
GPU Memory	Peak Memory (GB, Gigabytes)	5.70	5.99
	Additional QLoRA Memory (MB, Megabytes)	14.09	43.70
Training Performance	Average Batch Time (milliseconds)	990.44	1,048.09
	Average Time per Epoch (seconds)	343.7	357.12

Paged optimization refers to the systematic approach of managing and optimizing the paged attention mechanism. This includes tuning page sizes, managing page swapping strategies between GPU and CPU, and optimizing memory access patterns. This helps in taking up less space in the GPU and storing only what is required at that point in time from the CPU. The paged optimization results were remarkable: peak GPU memory reduced, which lowers the risk of OOM (Out-Of-Memory) errors during training, and additional QLoRA memory requirements decreased, meaning significantly less additional memory needed during fine-tuning, as shown in Table 10. The improvement in training performance may be due to the efficiency of the paging algorithm used to pre-fetch and cache data, ensuring that the GPU doesn't have to wait.

VI. CONCLUSION

In low resources scenario, where memory is at premium, this work wanted to experiment with tuning techniques LoRA and QLoRA. Detailed comparison with LoRA and QLoRA methods in terms of overall accuracy, GPU memory consumption, and training time has been performed and highlighted here. The work also reported and analyzed the results from different configurations of LoRA and QLoRA. Study results showed that LoRA and QLoRA techniques are good alternatives for full fine-tuning in resource-constrained environments, as the translation performance of the model fine-tuned using these techniques (except Idiomatic sentences) was similar to the performance of the full fine-tuned model, while significantly reducing resource usage. The choice between LoRA or QLoRA largely depends on the operational context. Since training is a one-time process its effect in the delay it adds to the operation of an organization would be less as compared to inference time. For organizations that would perform frequent fine-tuning of the model to be adapted quickly to updates, in this scenario, training time would be more important and the LoRA technique is advantageous due to its reduced training time. QLoRA is helpful only in cases of severe memory constraints.

Despite its contributions, this study has several limitations. Firstly, the findings are primarily based on a translation task, which may not generalize well to other NLP problems or domains where LoRA and QLoRA techniques might exhibit different performance. Furthermore, it was also subject to a certain dataset which restrict generalization on the datasets with different linguistic or domain characteristics. The translation performance particularly for idiomatic sentences were not good, exploration of combination of LoRA or QLoRA and partial fine-tuning of original parameters or a context-sensitive technique such as Retrieval Augmented Generation (RAG) is important.

There are many future directions that can open up important paths of research in enhancing the effectiveness and applicability of the LoRA and QLoRA techniques. A nice direction to explore is tuning parameters dynamically, where auto or adaptive techniques could be used for your hyperparameters during training, so there would be no tweaking needed from a human. Noise injection could be worth investigating to increase stability in fine-tuning in light of limited resources and reduce the risk of overfitting. Such directions could render LoRA and QLoRA even stronger and more applicable to different settings with limited resources.

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